



Passive learning and incidental exposure to news

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Abstract

Three criteria for political learning are opportunity, ability, and motivation. Models of passive learning call into question whether motivation is a necessary condition for learning about politics. Passive learning contends that incidental, or unintended, exposure to political content can produce learning among those unmotivated to view the content. To date, researchers have not found a way to conclusively test the idea of passive learning. By designing an experiment that gives individuals a choice in media content, we propose one strategy for testing passive learning among the unmotivated. Across four studies, the results illustrate limitations on the circumstances under which passive learning occurs and suggest that learning results from situations where incidental exposure sparks motivation.

Keywords: incidental exposure, passive learning, political knowledge

Introduction

How individuals come to understand political life is a central concern in a democracy. Possessing a store of correct political facts is seen as a critical prerequisite for connecting one's interests to one's political decisions and for participating in civic and political life (Delli Carpini & Keeter, 1996; Lau & Redlawsk, 1997; Lupia, 2016). The unequal distribution of political knowledge, coupled with attitudinal differences between those who are less and more informed, can be democratically consequential (Althaus, 2003). Yet, scholars disagree on the mechanisms that contribute to political learning. One research tradition argues that opportunity, ability, and motivation are necessary criteria for political learning (Delli Carpini & Keeter, 1996; Luskin, 1990). People must have the *opportunity* to encounter political information in order to learn. They must have the *ability*, including both the long-standing educational preparation and the short-term attentional capacity, to allow for political learning. Finally, they must have sufficient *motivation* in order to learn about politics.

Passive learning dismisses motivation as a necessary condition for learning about politics. Passive learning is the “acquisition of information by those with little interest in politics . . . This form of learning may be said to be largely passive in that *neither motivation nor interest* is a prerequisite for their attainment” (Zukin & Snyder, 1984, p. 629, emphasis added). Similarly, Tewksbury et al. (2001) note that “Passive learning, by definition, occurs in the *absence of a motivation* to become informed” (p. 534, emphasis added).

Passive learning involves two steps. First, one must encounter information that one was not motivated to seek. This has been called inadvertent exposure (Brundidge, 2010; Shehata et al., 2015), accidental exposure (Prior, 2005), preattentive exposure (Janiszewski, 1993), or incidental exposure (Baum,

2002; Fletcher & Nielsen, 2018; Gibson et al., 1994; Tewksbury et al., 2001; Yoo, 2009). Weeks and Lane (2020) note that “what makes exposure to news ‘incidental’ is whether or not individuals are motivated to seek it” (p. 1125). Second, one must gain knowledge, despite a lack of motivation for encountering the information. There is some ambiguity across definitions in terms of what motivation is necessary for passive learning, whether it is the motivation to learn, for example, or the motivation to seek information. In our experiments, we analyze what happens when people encounter content that they were not motivated to select and take up the question of other motivations in the discussion section.

Sorting out the evidence for passive learning is important given that several scholars proposed that its decline increased political polarization, reduced political knowledge, and depressed voter turnout (Baum & Kernell, 2009; Mutz, 2006; Prior, 2007). As the media environment allowed for greater choice, these scholars argued, the possibility for passive learning declined because incidental exposure became rarer. In an earlier broadcast era, politically uninterested audiences would watch television news and learn about politics because fewer alternative options existed. In a “post-broadcast era,” the politically uninterested could avoid news more effectively by switching to more entertaining content (Prior, 2007). Although declines in incidental exposure and passive learning may have characterized the era in which cable news emerged, social media and other digital forms of communication may be responsible for a resurgence (e.g., Bode, 2016; Feezell, 2018; Gil de Zúñiga et al., 2017; Lee & Kim, 2017; Oeldorf-Hirsch, 2018; Shehata & Strömbäck, 2018; Wojcieszak & Mutz, 2009).

Despite recent studies pointing to greater opportunities for passive learning, conclusive tests of passive learning remain elusive. Previous study designs cannot rule out the

explanation that learning occurred because some people were, in fact, motivated to learn. By creating an experimental design that gives individuals the ability to choose media content and then introduces political content to those who were not originally motivated to select it, we propose one strategy for testing the idea of passive learning that allows for a better separation of those motivated to seek content from those not motivated. Across four experiments, the results illustrate that learning can occur, but only under certain circumstances. The studies beg for a reconceptualization of passive learning, which we address in the discussion section.

Passive learning

To test the idea of passive learning from the news, one must demonstrate knowledge gains among those unmotivated to encounter news content. If some of those exposed to the information are, in fact, motivated to learn from it, then any documented gains in learning could be due to active, motivated learning, rather than passive, unmotivated learning. Several methods have been employed for studying passive learning, including field studies, surveys, and experiments. Although each makes progress, more work is needed to separate motivated from unmotivated participants.

Several studies analyzed passive learning on the basis of where people live (e.g., Delli Carpini et al., 1994; Zukin & Snyder, 1984). In a foundational study related to incidental exposure, Zukin and Snyder (1984) analyzed whether New Jersey residents' political knowledge differed depending on whether they lived in the part of the state that received New York radio and television or the part of the state that received Philadelphia radio and television. They found that New Jersey residents receiving New York media were able to name, without prompting and regardless of political interest, more New York mayoral candidates than New Jersey residents receiving Philadelphia media. There were few differences in these same groups' ability to name, without prompting, New Jersey gubernatorial candidates. Although Zukin and Snyder find knowledge differences among those expressing low interest in the outcome of the election, the study did not conclusively track other motivational mechanisms for knowledge gains (e.g., occupational or other personal motivations). This makes it difficult to conclude whether (a) political information was actively sought or accidentally stumbled upon and (b) whether knowledge gains were attributable to unmotivated learning, as passive learning would suggest, or active information seeking.

Survey research has explored passive learning by using self-report measures of media use, incidental exposure, and political interest. Tewksbury et al. (2001), for instance, used data from the Pew Research Center where participants were asked if they had "ever come across news when you may have been going online for a purpose other than to get the news?" In 1998, but not in 1996, the measure correlated with current affairs knowledge, even in the presence of a control for political interest. Using similar measures, some have found significant positive relationships between incidental exposure and knowledge (e.g., Marcinkowski & Doenović, 2021; Shehata et al., 2015; Tian & Robinson, 2009), while others have not (Anduiza et al., 2010). Gil de Zúñiga et al. (2021) identified a negative relationship between incidental exposure and political knowledge, yet also showed that incidental exposure makes a positive contribution to political knowledge through

engagement and cognitive elaboration. With respect to social media, where scholars have proposed that news is encountered incidentally, some find non-significant, or even negative, relationships between Facebook and Twitter use and political knowledge (Bode, 2016; Cacciatore et al., 2018; Oeldorf-Hirsch, 2018). A novel combination of survey data and data obtained by scraping users' Facebook News feeds also uncovered no association between news-related social media content and political knowledge (Wells & Thorson, 2017).

Although some survey-based studies provide evidence favoring passive learning (e.g., Shehata et al., 2015; Tewksbury et al., 2001; Tian & Robinson, 2009), alternative explanations are possible. First, in studies using questions asking directly about incidental exposure, it is not clear whether people can accurately recall and report this behavior. As Palmgreen (1984) wrote, "there is controversy over the extent to which respondents are conscious of their motives for consumption or can be made conscious of such motives through closed- or open-ended techniques" (p. 23). Indeed, Tewksbury et al. (2001) concede that their self-report measure is "less than ideal" (p. 546). Second, and more important, it is *still possible* that the knowledge gains came from purposeful—as opposed to incidental—learning. Those reporting more *purposeful* selection of news also report more *accidental* exposure to news (Oeldorf-Hirsch, 2018; Tewksbury et al., 2001), adding weight to this alternative argument. As with all survey studies, it is not possible to measure or control for all of the reasons that individuals may have been motivated to choose particular content. Without the benefit of experimental control, causal relationships cannot be uncovered.

Few experiments have tested the idea of passive learning. Studies of advertising effectiveness and digital browsing behaviors have made some strides, however. Shapiro et al. (1997), for instance, analyzed the effects of incidental exposure by randomly assigning respondents to read an article that either did or did not have advertisements alongside the main article text. To try to simulate unmotivated exposure, Shapiro and colleagues asked participants to keep their cursor in the spaces between words while reading through the article. They found that buying intentions for the advertised products were higher among study participants reading the article with the accompanying advertisements than those reading the article without the ads. Other work in advertising has used similar methods (McQuarrie & Mick, 2003; Yoo, 2009). In a notable extension, Lee (2009) analyzed whether incidental exposure would affect study participants' ratings of issue importance. Lee randomly assigned study subjects to browse a website that either did or did not contain banner ads with links to environmental news stories. Results showed little effect of the banner ads on ratings of issue importance unless study participants received explicit instructions to click on the ads. An extension of this work found increases in knowledge for those randomly assigned to view a university website containing banner ads linking to news stories compared to those randomly assigned to view the same university website without the banner ads (Lee & Kim, 2017). Nanz and Matthes (2020) manipulated whether, in two of seven article hyperlinks in a list, a headline mentioned a nearby or far-away city. People recalled the headline about the far-away city at rates greater than chance, suggestive of passive learning. Yet, as the authors note, the results are not definitive, as people could have had reasons to attend to the headline about the far-off city.

Recent experimental work assessing news exposure via social media also illustrates the necessity for distinguishing between motivated and unmotivated exposure. Bode (2016) found greater knowledge among those randomly assigned to view a Facebook Feed containing political content compared to those viewing a Feed lacking this content. Building upon this research, Feezell (2018) randomly assigned study subjects either to participate in a treatment Facebook group that received political news stories or a control Facebook group that did not. Those seeing the political stories were more likely to report that the issues contained therein were important, particularly those with lower political interest. Anspach et al. (2019) found modest increases in knowledge when comparing a control condition where respondents did not see any stimuli to a condition where respondents viewed a mock Facebook News Feed that contained several article previews, including one about the news item of interest.

These experimental studies make progress in understanding passive learning, yet more work is needed to distinguish between motivated and unmotivated exposure. In each study, people could have been motivated to attend to the ads or political content because they found it more interesting than the other content made available. Measuring political interest as a moderator helps to assess whether those uninterested are also affected, but political interest only captures one type of motivation; people may have other reasons to attend to content (e.g., a political article referencing one's hometown may be chosen, even if the person reports little political interest). These ambiguities make it difficult to sort out whether any learning that occurred was passive or active. What is needed is a way to isolate learning among those unmotivated to seek political content.

Overall, some studies have found evidence of passive learning, while others suggest mixed or non-existent knowledge outcomes. Regardless of the method employed, however, questions remain about whether learning accrues among those who are motivated to obtain information or whether there is evidence of passive learning. Based on the passive learning model, the key hypothesis that we investigate across four studies is that *those incidentally exposed to news will know more about the topics covered in the news compared to those not incidentally exposed to news*.

Given the critiques associated with prior methods, it is vital to employ a new design. To study passive learning in a way that more effectively separates motivated and unmotivated news seekers, we adapt an experimental methodology that incorporates choice into the design. Several projects have incorporated choice into experiments (e.g., Arceneaux & Johnson, 2013; Druckman et al., 2012; Levendusky, 2013; Prior, 2007; Stroud et al., 2019). To date, these novel experimental designs have not been used to analyze incidental exposure. By building on these designs, we propose a new approach that allows us to test passive learning by more clearly separating those motivated to seek out news from those preferring other options.

Study 1 Radio breaks: Method

Study 1 asked participants to look at a website designed to resemble a radio. In this experiment, participants were randomly assigned to one of three radio choice conditions. In the *Music-Only* condition, participants were free to choose among six non-news, music radio stations (e.g., Country,

Top 40). This condition provides a baseline level of knowledge among those not exposed to any news content. In the *Music-News* condition, participants were free to choose among five non-news, music stations, and one local news station. This allows for an evaluation of learning among those who are motivated to *choose* news when given the option. In the *News Break* condition, our incidental exposure condition, participants were again able to choose among five non-news, music stations, and one local news station. At exactly 2 min, the station to which the respondent was listening switched to a 2-min news break. For example, a participant listening to the Classical music station would hear the current song end and would hear a reporter with music in the background announce "And now the news..." followed by a 2-min news clip about the city mayor, manager, and council. This would occur only on the station to which the respondent was listening. If the respondent switched to a different station, all of the other stations would continue to play music, or news unrelated to the two-minute break. Respondents in all conditions were asked to spend 5 min listening to the radio website. Stations for the radio website were selected based on the top local radio stations and genres as determined by Arbitron. Respondent listening behavior was tracked unobtrusively using the MediaLab program.

By comparing the *Music-News* and *News Break* conditions, we can understand whether passive learning takes place, holding constant learning among those who were motivated to choose the news in the first place. Differences between these conditions would indicate that incidental exposure increases knowledge *above and beyond* those who choose news.

Participants

Participants were U.S. residents aged 18 years and older and were recruited from a local library, the student union on a large southwestern U.S. university campus, and a local coffee shop. Participants were offered a small cash incentive for their participation. In total, 232 participants were randomly assigned to one of the three previously described conditions.¹ Fifty-four percent of the study participants were male. Thirty-three percent had completed a high school education or less, 38% had completed more than a high school education, but had not attained a 4-year college degree, and 29% had a 4-year college degree or more. Respondents were 30.47 years of age on average ($SD = 12.42$). Thirty-five percent identified as Black, 33% identified as White, 24% as Hispanic, and 14% as Asian. Chi-square and analysis of variance (ANOVA) tests confirmed that there were no differences among the conditions on these demographic variables.

Measures

After engaging with the website, respondents were asked five local political knowledge questions that could be learned from the news break (e.g., What is the name of the mayor? Full question wording included in the [Supplementary Appendix](#)). Correct responses were scored as 1 and incorrect responses as 0. Summing correct responses, respondents answered an average of 1.21 questions correctly ($SD = 1.21$).²

Pre-test

Before beginning the study, we conducted a pre-test to evaluate whether people were better able to answer the knowledge questions after hearing the news break. We wanted to ensure that people could, in fact, learn the facts we included in our

knowledge question battery from the news segment that we selected. A convenience sample of 44 U.S. residents aged 18 years and older completed the pre-test. Participants were randomly assigned either to first hear the news radio segment and then to answer the knowledge questions or to first answer the knowledge questions and then to hear the news radio segment.

The results provided consistent evidence that the news segment enhanced knowledge. For each of the five local knowledge questions, the percentage of correct answers was higher for those who had listened to the news segment compared to those who had not. Summing correct answers to the questions, those who listened to the news segment first and then answered the knowledge questions had significantly higher scores compared to those who answered the questions prior to hearing the segment (news first $M = 3.28$, $SD = 1.28$; knowledge first $M = 2.11$, $SD = 1.52$; $t(42) = 2.78$, $p < .01$). As people *could* learn the information contained in our knowledge questions from the segment, we proceeded with our study.

Study 1 Radio breaks: Results

We first analyzed how subjects reacted at the 2-min mark by condition. The news segment of interest began in the *News Break* condition at 2 min and ended at 4 min. We analyzed how long subjects spent with the station at the 2-min mark before they switched to another station. Across the conditions, respondents spent an average of 64.32 s with a station ($SE = 2.95$) between 120 and 240 s of their listening experience. There was a significant main effect of condition on time spent with the station to which one was listening at the 2-min mark ($F(2, 229) = 6.76$, $p < .01$). Examining the differences among the conditions, respondents in the *News Break* condition spent significantly less time with the station playing at 2 min ($M = 50.06$, $SE = 4.93$) than those in the *Music-Only* condition ($M = 68.36$, $SE = 5.06$, $p < .05$) and those in the *Music-News* condition ($M = 74.89$, $SE = 4.97$, $p < .01$). As anticipated, there were no significant differences between the *Music-Only* and *Music-News* conditions ($p = .36$).³

Turning to the local knowledge measures, subjects in the *Music-Only* condition answered an average of 1.16 questions out of the five local political knowledge questions correctly ($SE = 0.14$), subjects in the *Music-News* condition answered an average of 1.08 questions correctly ($SE = 0.14$), and subjects in the *News Break* condition answered an average of 1.39 questions correctly ($SE = 0.14$). Although the results trend in the expected direction, an ANOVA reveals no significant differences ($F(2, 229) = 1.45$, $p = .24$).⁴

Given that we found no significant differences in local knowledge, we further analyzed whether those listening to the news answered more of the knowledge questions correctly to test whether the stimuli were working as intended. As shown in Table 1, the more time that participants purposefully spent with the news—a possibility in the *News Break* and *Music-News* conditions—the more knowledgeable they were. Furthermore, the more time that respondents spent with the news break in the *News Break* condition, even after taking into account their purposeful news selection, the more knowledgeable they were.

Study 1 Radio breaks: Discussion

The results of Study 1 provide limited evidence of passive learning, particularly not in an environment that provides

Table 1. Study 1: Regression analysis of local political knowledge

Unstandardized coefficients	
Condition	
<i>Music-News</i>	−0.24 (0.19)
<i>News Break</i>	−0.46 (0.26)
Time with News Break (<i>News Break</i> condition, otherwise 0)	0.01* (0.003)
Time with News Station (<i>Music-News</i> and <i>News Break</i> conditions, otherwise 0)	0.01** (0.001)
Constant	1.16** (0.13)
R^2	0.10

Note: Music-Only is the excluded condition; * $p < .05$, ** $p < .01$.

citizens with the ability to choose something else. When confronted with a news break in the middle of listening to music, many participants responded by finding another station. There were no differences across the conditions in knowledge about the political facts included in the news segment. Those who did stay with the news when it began unexpectedly and those who purposefully chose the news did know more about local politics, yet they were few in number and did not result in any aggregate differences across the conditions.

The lack of differences in local political knowledge between the *Music-Only*, *Music-News*, and *News Break* conditions could be attributed to many different factors, some more plausible than others. First, this lack of difference could be the result of inadequate statistical power. Following Murphy et al. (2008), for a sample of this size, power is 0.80 for rejecting the null hypothesis if the randomly assigned conditions account for 4% of the variance in knowledge. Considering this result, only small effects would be missed by the present analysis. In other words, if there is a passive learning effect, it would be notably small.

Second, the lack of a difference could be due to something unique about the news segment we selected for this study. It could be that this particular news segment does not inspire political learning. Yet, the pre-test results show that people *could* learn from the segment if they decided to listen to it, but many elected not to. Our subsequent analysis also shows that those engaging with the news segments learned more (Table 1), but the cross-sectional nature of the analysis does not provide irrefutable evidence of passive learning. Study 2 tests the same hypothesis in a different experimental context with different news content.

Study 2 Internet sidebars: Method

Study 2 analyzed passive learning based on exposure to an entertainment-focused article with news included in the sidebar. Study participants were randomly assigned to one of four different conditions. The choices available to respondents in the first three conditions resembled those from Study 1. In the *Entertainment-Only* condition, participants were given five entertainment articles from which to choose (e.g., articles about travel or a Mediterranean diet). In the *Entertainment-News* condition, participants were given four entertainment articles and one news article from which to choose. The news article

discussed important upcoming political events, such as the election and the president's State of the Union address. In the *News Sidebar* condition, which introduced incidental exposure, participants were given the same four entertainment articles and one news article from which to choose as they were given in the *Entertainment-News* condition; however, the entertainment articles also contained a news sidebar that summarized the relevant political news from the news article. As in Study 1, comparing the *News Sidebar* and *Entertainment-News* conditions allows us to isolate learning among those not motivated to seek news. For the first three conditions, participants were asked to select one article from the available choices. We added a condition that was not included in Study 1 here. In the *News-Only* condition, respondents were forced to read the news article and were not given any choice among articles.

Participants

Three hundred participants completed the second study via Amazon.com's Mechanical Turk. Study participants were U.S. residents aged 18 years and older, 52% female, 84% White, 10% Black, 4% Hispanic, and 3% Asian. Fourteen percent had a high school education or less, 38% had completed more than a high school education, but had not attained a 4-year college degree, and 48% had a 4-year college degree or more. Respondents were 38.18 years of age on average ($SD = 12.97$). Chi-square tests and ANOVAs confirmed that there were no differences across the conditions on these variables.

Measures

Our main dependent variable consisted of six knowledge questions where the answers could be found in the news article (contained in the *News-Only*, *Entertainment-News*, and *News Sidebar* conditions) or in the news sidebar running alongside the entertainment articles in the *News Sidebar* condition. Across all conditions, respondents answered an average of 3.46 questions correctly ($SD = 1.52$).

Study 2 Internet sidebars: Results

Knowledge connected to the news differed significantly across the conditions ($F(3, 296) = 12.72, p < .01$). The *Entertainment-Only* condition ($M = 2.78, SE = 0.16$) resulted in significantly lower knowledge compared to the three other conditions ($p < .01$). Further, the *News-Only* condition ($M = 4.22, SE = 0.17$) resulted in significantly greater knowledge compared to the three other conditions ($p < .01$). These results demonstrate that people can learn from the news when forced to view it. The critical test for passive learning is the difference between the *Entertainment-News* condition ($M = 3.49, SE = 0.17$) and the *News Sidebar* condition ($M = 3.40, SE = 0.16$). Here, the difference was not significant ($p = .68$).⁵

Study 2 Internet sidebars: Discussion

Evidence from Study 2 did not favor the passive learning hypothesis. If the passive learning hypothesis were correct, we would have anticipated that those in the *News Sidebar* condition would have answered more knowledge questions correctly than those in the *Entertainment-News* condition. There were no significant differences between these conditions. This adds weight to the finding from Study 1, where passive

learning was minimal. Study 2 extends the results of Study 1 by analyzing a different medium (radio for Study 1, Internet for Study 2) and by including the *News-Only* condition that allowed us to demonstrate that it was possible for people to learn from the news if they were forced to view it. Both Study 1 and Study 2 gave people a great deal of choice. In Study 1, participants were able to switch to a different radio station at any point. In Study 2, participants were able to move on to the next part of the survey whenever they wanted. In effect, we made it easy for respondents to avoid the news sidebar when it was available. Yet it is possible that the sidebar does not facilitate learning. The *News-Only* condition required participants to look at the entire article, not a page with the sidebar. This makes it unclear whether people *could* learn from a news sidebar. In Study 3, we constrain study participants' choices by requiring them to spend 30 s with an article they choose. In this situation, we analyze whether we can create circumstances in which there is evidence for passive learning. Further, we check whether experimental subjects can learn from a news sidebar when they are forced to stay on the page.

Study 3 Internet sidebars with forced time: Method

Study 3 employed a framework similar to the one described in Study 2. The conditions were identical to Study 2, with two exceptions. First, we used a different news story. In the *News-Only*, *Entertainment-News*, and *News Sidebar* conditions, the news described Takoma Park, Maryland extending voting rights to 16- and 17-year-olds in municipal elections. Second, rather than giving participants full discretion to continue with the study at any time as we did in Study 2, we required participants to stay on the article page they selected for 30 s. This notable departure examines passive learning in situations with some constraints. If we force people to stay on a page for just 30 s, will we see evidence of learning among those who did not choose the voting rights article, but could read about it in the article sidebar?

Participants

Three hundred and twenty-nine participants completed the third study via Amazon.com's Mechanical Turk. Study participants were U.S. residents aged 18 years and older, 49% female, 82% White, 10% Asian, 5% Hispanic, and 5% Black. Eleven percent had a high school education or less, 43% had completed more than a high school education, but had not attained a 4-year college degree, and 46% had a 4-year college degree or more. Respondents were 33.50 years of age on average ($SD = 12.50$). Chi-square tests and ANOVAs confirmed that there were no differences between the conditions on these variables.

Measures

To measure our main dependent variable, respondents were asked a series of five knowledge questions where the answers could be found in the news article (contained in the *News-Only*, *Entertainment-News*, and *News Sidebar* conditions) and the sidebar running alongside the entertainment articles in the *News Sidebar* condition. Summing correct responses, respondents answered an average of 2.08 questions correctly ($SD = 1.48$).

Study 3 Internet sidebars with forced time: Results

There were significant differences across conditions in knowledge that could be ascertained from news exposure ($F(3, 325) = 58.66, p < .01$). As in Study 2, those in the *News-Only* condition ($M = 3.46, SE = 0.13$) answered more knowledge questions correctly compared to those in the other three conditions ($p < .01$). Further, those in the *Entertainment-Only* condition ($M = 1.22, SE = 0.13$) answered fewer questions correctly in comparison to those in the *News Sidebar* condition ($M = 2.25, SE = 0.13, p < .01$). Differing from Study 2, there was not a significant difference in the number of knowledge items answered correctly for those in the *Entertainment-Only* condition and those in the *Entertainment-News* condition ($M = 1.46, SE = 0.13, p = .20$). In fact, only 10% of those assigned to the *Entertainment-News* condition chose the article about minors voting.⁶ The critical test of passive learning—a comparison between the *Entertainment-News* condition and the *News Sidebar* condition—was significant ($p < .01$). Those in the *News Sidebar* condition answered significantly more knowledge questions correctly compared to those in the *Entertainment-News* condition.

Study 3 Internet sidebars with forced time: Discussion

Study 3 provides some evidence of passive learning when participant choice is constrained even for a short amount of time (30 s). In this situation, those in the *News Sidebar* condition answered significantly more knowledge questions correctly compared to those in the *Entertainment-News* condition. This also provides evidence that subjects in Study 2 *could* have learned from the news sidebar had they been motivated to attend to it. Whereas Study 3 demonstrates passive learning when people are forced to stay on a page, we wanted to examine the potential for passive learning in more natural circumstances. Study 4 examines the potential for passive learning in the context of social media.

Study 4 Social media with news ledes: Method

We conducted an experiment with conditions closely resembling those in the prior studies, this time conducted using a mock Facebook News Feed interface. The interface allowed for interactive engagement with content (e.g., clicking on, liking, and sharing articles; leaving a comment), but was not connected to study participants' actual Facebook page.

Participants were randomly assigned to one of four conditions. The first condition, *Entertainment-Only*, showed participants a Facebook News Feed with five headlines about entertainment topics such as sports, movies, and travel. Respondents were able to choose whatever they wanted to read and could continue at any time. In the second condition, *Entertainment-News*, participants were free to choose among four entertainment headlines and one news headline. This allowed for an evaluation of learning among those who *choose* news when given the option. The news headline described how Rhode Island began offering its prison inmates medications approved to treat opioid addiction. In the third condition, *Entertainment-News with Ledes*, participants were again able to choose among four entertainment headlines and

one news headline. This condition, however, also introduced the possibility of incidental exposure. All of the articles were accompanied by ledes that included some information about the articles. In this way, even those not interested enough to click on the article could still be incidentally exposed to key information from it. The articles were randomly ordered. The fourth condition, *News-Only*, showed people the news article without any choice.⁷

By comparing the *Entertainment-News* and *Entertainment-News with Ledes* conditions, we can understand whether passive learning takes place, holding constant learning among those who were motivated to choose the news in the first place. Differences between these conditions would indicate that incidental exposure increases knowledge above and beyond those who would choose news. The *Entertainment-Only* and *News-Only* conditions allow us to understand baseline knowledge and “full” knowledge based on forced exposure to the news item.

Participants

Data from 526 U.S. residents aged 18 years and above were gathered from Dynata. Fifty-nine percent identified as female, 64% as White, 31% as Hispanic, 21% as Black, and 3% as Asian. Twenty-two percent had a high school degree or less, 50% had completed more than a high school education, but had not attained a 4-year college degree, and 28% had a 4-year college degree or more. The average age was 45.77 ($SD = 19.13$). Chi-square tests and ANOVAs confirmed that there were no differences among the conditions.

Measures

The key dependent variable was how much respondents knew about the news article. We asked respondents five questions that could be ascertained from reading the lede of the article appearing alongside the Facebook post in the *Entertainment-News with Ledes* condition or from reading the full article in the *Entertainment-News*, *Entertainment-News with Ledes*, or *News* conditions. We summed the total number of correct responses ($M = 2.12, SD = 1.41$).

Study 4 Social media with news ledes: Results

There were significant differences in knowledge across the conditions ($F(3, 522) = 67.99, p < .01$). As predicted, those in the *Entertainment-Only* condition had the lowest levels of knowledge ($M = 1.50, SE = 0.11$), although not significantly different from those in *Entertainment-News* condition ($M = 1.64, SE = 0.10, p = .34$). Those in the *Entertainment-News with Ledes* condition ($M = 1.98, SE = 0.10$) had significantly greater knowledge than those in the *Entertainment-Only* ($p < .01$) and *Entertainment-News* ($p < .05$) conditions. Finally, those in the *News* condition had significantly greater knowledge than those in the other three conditions ($M = 3.41, SE = 0.11, p < .01$).

Study 4 Social media with news ledes: Discussion

The results of this study demonstrate that passive learning can occur. Participants who were incidentally exposed to the article lede (*Entertainment-News with Ledes*) were more knowledgeable than those who did not see the lede (*Entertainment* and *Entertainment-News*). We note that participants who

had the option of reading the news article (*Entertainment-News* condition) were no more knowledgeable compared to those who did not have the option (*Entertainment* condition), suggesting that this article did not inspire many people to read it compared to the others in the choice set.⁸

General discussion

There is evidence consistent with the passive learning hypothesis in some, but not all, of the studies we fielded (see a summary of findings in Table 2). The pattern is telling. Theorists propose that the necessary requirements for political learning are opportunity, ability, and motivation (Delli Carpini & Keeter, 1996; Lau & Redlawsk, 1997; Lupia, 2016). Passive learning suggests that learning can occur absent motivation (Tewksbury et al., 2001; Weeks & Lane, 2020; Zukin & Snyder, 1984), leaving two criteria: learning should occur if the *opportunity* arises and people have the *ability* to do so. Across our four studies, those in the incidental exposure conditions had the opportunity to learn and our evidence supports the idea that they had the ability to learn. Study participants had the opportunity to learn because the information was present, whether in a radio news break, an Internet sidebar, or a Facebook post that provided details about the news article. We also have evidence that our study participants had the ability to learn. Those who were forced to engage with the news in the *News-Only* condition in Studies 2–4 knew more than those in the other conditions and although we did not have this condition in Study 1, our pretest showed similar results. Further, Study 3 forced participants to stay on the page with a news sidebar for 30 s, whereas Study 2 allowed participants to leave the page at their leisure. Incidentally exposed study participants in Study 3, but not in Study 2, learned more than those who did not have the opportunity for incidental exposure.⁹ These results suggest that people had the ability to learn. Even though participants across four studies had the opportunity and ability to learn, they did not consistently do so. If passive learning were correct, we would have anticipated consistently higher knowledge in the incidental exposure conditions (Table 2, column c) compared to the conditions with an identical choice set, but without incidental exposure (Table 2, column b). Across these studies, however, we see evidence consistent with passive learning in Studies 3 and 4, but not in Study 2 and only limited evidence in Study 1 where the main effect was not significant, but time with the news break predicted local political knowledge.

Passive learning, as conceptualized in previous research, cannot account for the results across the four studies. Something beyond opportunity and ability must be at play to

explain the pattern. Contrary to the passive learning definition, we propose that motivations influence whether people learn—even when they are incidentally exposed. This idea offers a parsimonious explanation for the patterns uncovered. In Study 1, motivation can explain the absence of an effect. In a high choice environment, those who were uninterested in hearing the news changed the radio channel. They were motivated by their disinterest to select other content. In Study 2, people ignored the news sidebar because the content did not motivate their attention whereas in Study 3, participants were motivated by a time limit to wait on the page, and they learned. And in Study 4, people found the content interesting enough that they read the news lede, even though they were not motivated to click and read the article. Motivations can explain our findings whereas passive learning is unable to account for the variation across studies.

In our experiments, we manipulated whether those who did not select news were exposed nonetheless, thus isolating the motivation to select content. One might also consider the motivation to process content (e.g., Matthes et al., 2020). These two motivations, seeking and processing content, are likely correlated, as someone who selects news is likely motivated to process it. Our studies explore what happens when a person not motivated to seek content encounters it anyway. Our results suggest that whether learning results after unintended exposure is a question of whether the content is compelling enough to activate a motivation to process it.

The studies also raise questions about whether there are any circumstances under which passive learning occurs. We hold out the possibility that some low level of processing could occur from unintended exposure absent a processing motivation, but it would be a form of subliminal processing. This seems unlikely to produce any form of political learning like we analyze here, and certainly would not yield the sort of factual knowledge required of an informed public. In fact, research suggests that subliminal processing yields no reportability (Dehaene et al., 2006). Here we concur with Matthes et al. (2020) and Wieland and Kleinen-von Königslöw (2020) that any learning resulting from truly unmotivated exposure would be limited at best.

Which motivations affect learning from incidentally encountered news is ripe for additional analysis. Although numerous motivations have been identified, one categorization scheme seems particularly relevant: the distinction between intrinsic and extrinsic motivation (see Lepper & Henderlong, 2000). Intrinsic motivations occur when people are motivated by an activity itself; that is, when they find the activity inherently engaging. Extrinsic motivations occur when people engage in actions because of an external reward or punishment. Both types of motivations seem to exist. Forcing people to

Table 2. Knowledge scores across studies

	Mean (SE)			
	a. Entertainment-Only Choices	b. Entertainment + News Choices	c. Entertainment + News Choices + Incidental Exposure	d. News-Only
Study 1—Radio News Breaks	1.16 (0.14) ^a	1.08 (0.14) ^a	1.39 (0.14) ^a	NA
Study 2—Internet Sidebars	2.78 (0.16) ^a	3.49 (0.17) ^b	3.40 (0.16) ^b	4.22 (0.17) ^c
Study 3—Internet Sidebars with Forced Time	1.22 (0.13) ^a	1.46 (0.13) ^a	2.25 (0.13) ^b	3.46 (0.13) ^c
Study 4—Social Media with News Ledes	1.50 (0.11) ^a	1.64 (0.10) ^a	1.98 (0.10) ^b	3.41 (0.11) ^c

Note: Different superscripts indicate significant differences within each row. NA indicates that this condition was not included in the study.

attend to news content (the Study 2–4 *News-Only* conditions and the Study 3 requirement to remain on the page for 30 s) seems more closely aligned with an extrinsic motivation. People engaged in the behavior because the experimental situation required it of them. When this occurs, learning can result. Extrinsic motivations also could result based on the other options available—if the country music radio station were the only one of interest to the participant in Study 1, for example, then waiting through a news break is arguably a better, or similar, option to switching the station. Another extrinsic motivation could be the effort required to change the station or click to another article—the effort required may be sufficient to keep people on a station. The same logic underlies inheritance effects where people stay tuned to a television channel instead of looking for another option when the programming changes (Webster, 2006).

The possibility of intrinsic motivations also exists. For some subset of the study participants, the available news may have been compelling enough to hold their attention, even though they were not interested in the first place. This coincides with Matthes et al.'s (2020) notion of relevance, whereby people determine whether a piece of content is worthy of additional information processing. For some, the news to which they were incidentally exposed was deemed worth attention, and they subsequently gained knowledge. The instances in which incidental exposure produced learning here (e.g., Study 4) can serve as a starting point for exploring what yields attraction to news. Indeed, we believe that this is the most important step for future research: creating and testing theories about what sparks attention to content that one is unmotivated to select. For example, one can evaluate whether the mix of available choices affects learning; more news options, for instance, may increase learning if they put people in a state-of-mind more receptive to processing news. It also will be valuable to test how the results vary by individual characteristics. In our data, for example, incidental exposure results in greater knowledge gains among those with higher pre-existing levels of background political knowledge, a finding consistent with the opportunity–ability–motivation model.¹⁰

In addition to theoretical contributions, the studies make important methodological advancements. Past passive learning studies have contrasted a condition where some study subjects are exposed to media content, whether news from a competing market (Zukin & Snyder, 1984), advertisements (Shapiro et al., 1997), or banner ads containing news (Lee, 2009), among others. Although these studies help us to understand situations in which people can learn, the study designs do not allow for a conclusive separation between those motivated to learn the information and those not motivated. This is because all previous studies either (a) were not experimental or (b) did not incorporate a strategy that clearly separated those motivated from those unmotivated to select the content. Survey-based studies measure and control for motivations, but we cannot know whether people were motivated for reasons unaddressed in the questionnaires. For instance, there are legitimate reasons that someone with little political interest still may be motivated to learn about politics (e.g., career goals, social/interpersonal concerns). Experimental studies that do not incorporate choice make it difficult to tell whether experimental subjects are genuinely interested in the material designed to test the effects of incidental exposure. Study participants forced to read an uninteresting article in a lab setting

with an attractive ad alongside the article may be motivated to spend time with the ad. These critical difficulties in assessing passive learning, we propose, can be overcome by building choice into experimental design (e.g., Arceneaux & Johnson, 2013; Druckman et al., 2012; Levendusky, 2013; Stroud et al., 2019).

In this study, we created different choice conditions—in some conditions, participants did not have the option to attend to the news, while in others they did. In the critical incidental exposure conditions (the *News Break*, *News Sidebar*, and *Entertainment-News with Ledes* conditions), participants uninterested in news were exposed nonetheless. This new twist on experimental design provides a stronger way to test the passive learning hypothesis. Read in the context of the present research design, past passive learning studies compared (a) those exposed only to entertainment to (b) those with the ability to choose news *and* those incidentally exposed with controls for motivations that may have prompted people to choose the news. Because of the conflation of factors, tracking (un)motivated learning was difficult. With our design, we can more closely capture the situation in which passive learning could occur by parsing out those uninterested in news based on their expressed choices. This is not to say that our design is flawless. It is possible that some subset of respondents would have been equally happy with the news or their chosen entertainment option, for example. Nonetheless, their active selection of non-news content suggests that they were less motivated to seek out news than other choices.

Additional limitations are notable. First, although we underwent extensive efforts to create realistic choice settings, the stimuli and browsing experience were still artificial. Efforts to make the choice setting more realistic may have to relinquish some of the experimental control which was critical to our study. Nonetheless, efforts to evaluate these findings in real-world settings are important. Second, we examined short-term knowledge gains, without assessing the more long-term knowledge requirements proposed by democratic theorists (Delli Carpini & Keeter, 1996; Lau & Redlawsk, 1997; Lupia, 2016). Whether incidental exposure has a lasting impact on learning requires more research. The short-term changes we document, however, represent an important first step. Third, we provide a motivational explanation for our findings, but we did not explicitly measure, or manipulate, motivations. The motivational account is consistent with past work on political learning (Delli Carpini & Keeter, 1996; Luskin, 1990), making it a plausible explanation. Moreover, motivations are implicitly incorporated in the study because they govern where people choose to allocate their attention. Experimentally manipulating motivations is a possible strategy, but experimental manipulations are rarely effective on all study subjects, again leaving open the possibility that content intended to be encountered incidentally is purposefully selected over other experimentally available content.

The studies and discussion here raise important questions about passive learning. In particular, the idea that incidental exposure will produce passive learning of substantive facts seems suspect. We propose incidental exposure *can* give rise to learning, but these circumstances likely arise when the content or the situation *motivates* people to process further, in line with models from Matthes et al. (2020) and Wieland and Kleinen-von Königslöw (2020). This discussion of motivations, however, is a notable departure from the idea of passive learning, which is defined as learning *absent* motivation. We

conclude that the traditional conception of passive learning needs to be re-thought. Much research that has been done under the label of passive learning, we contend, is actually about motivated learning. A distinct concept, perhaps incidental learning from Wieland and Kleinen-von Königslöw (2020), seems apropos.

The real question is what circumstances motivate information processing. Our studies provide empirical fuel to Thorson's (2020) suggestion that we need to spend more time researching what attracts people to news—the motivational component that can yield learning. Increasing political knowledge involves more than merely pushing news onto the public. It requires identifying ways to encourage citizens to remain attentive after encountering news.

Supplementary material

Supplementary material is available at *Journal of Communication* online.

Data availability statement

The data underlying this article will be shared on reasonable request to the corresponding author.

Conflict of Interest: none declared.

Funding

We are grateful for support for this research from the Moody College of Communication, Democracy Fund, and John S. and James L. Knight Foundation.

Acknowledgments

The authors thank Kaylinh Ly, Sameer Bhuchar, Jordan Humphreys, and Leilani Kelley for their assistance with data collection.

Notes

1. Data for an additional 26 participants were removed from the dataset because they encountered computer problems.
2. The data underlying this article will be shared on reasonable request to the corresponding author.
3. Average time spent with the news channel did not differ between the *Music-News* ($M = 27.05$, $SE = 6.14$) and *News Break* conditions ($M = 46.87$, $SE = 8.51$; $t(141.56) = 1.89$, $p = .06$). As the result approaches significance, however, we repeated our analyses with a control for time with news. The conclusions were unchanged.
4. To check for hypothesis guessing, 32 participants assigned to the *News Break* condition were asked, at the end of the study, to say what they thought the study purpose was, when they reached this conclusion, and whether their thoughts about the study purpose affected their behavior in any way. Only two participants mentioned the news break and both said that they reached this conclusion only after they had finished browsing the radio website. This yields little evidence of hypothesis guessing.
5. The percentage of respondents choosing news in the *Entertainment-News* and *News Sidebar* conditions was not significantly different ($z = 1.60$, $p = .12$).
6. The percentage of respondents choosing news in the *Entertainment-News* and *News Sidebar* conditions was not significantly different ($z = 0.54$, $p = .59$).

7. We had two other conditions, but they did not yield any significant findings and involve theorizing beyond the bounds of this article. The first was an *Entertainment-News with Social Endorsements* condition, which included the same stimuli as the *Entertainment-News* condition, but added social endorsements in the form of Facebook "Likes," with the news article receiving between 95 and 105 likes and the other articles receiving fewer than 10 likes. The second was *Entertainment-News with Ledes and Social Endorsements*, which included the same stimuli as the *Entertainment-News with Ledes* condition, but added the same social endorsements. Results showed no differences between the conditions with social endorsements and those without, so we do not discuss these conditions further.
8. The percentage of respondents clicking news in the *Entertainment-News* and *Entertainment-News with Ledes* conditions was not significantly different ($z = 1.63$, $p = .10$).
9. The stimuli change across these two studies is a possible alternative explanation—but even here, there would have to be something more compelling about one stimuli compared to another which points to something beyond opportunity and ability.
10. We tested whether a measure of background political knowledge, such as knowing which political party is more conservative on the national level, moderated the effect of the experimental conditions on the knowledge dependent variable. The measure, available in Studies 2–4, was not a significant moderator in Study 2. In Studies 3 and 4, incidental exposure produced greater learning among those with greater background knowledge. See more detail in the [Supplementary Appendix](#).

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